

Lunar Helium-3 Fusion Under CVP Control: A Governance-First Overlay for Aneutronic Confinement and Direct Energy Conversion

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Abstract

Lunar fusion strategies often target Helium-3 (He-3), an isotope implanted into lunar regolith by solar wind over geologic time. He-3 reactions (e.g., $D + He-3$) are largely aneutronic relative to $D-T$, producing a higher fraction of charged products and enabling direct electromagnetic energy conversion. The tradeoff is severe: He-3 operation demands higher temperature and stronger confinement, where turbulence-driven transport, edge drift, and MHD instability dominate the loss channels. This paper proposes the CVP Confinement Layer as a governance-first control overlay for He-3 fusion—an invariant-driven state layer, curvature-response actuation, strict local recovery, and TEVR (Test, Evaluation, Verification, Replication) proof bundles—intended to turn instability suppression into a repeatable engineering outcome rather than a one-off operating point.

1. Motivation: Why Lunar He-3 Becomes a Control Problem

The lunar He-3 proposition is attractive because charged reaction products can be converted directly to electricity via electromagnetic fields, bypassing a thermal steam cycle. However, achieving that advantage requires operating regimes where confinement is routinely lost to turbulence and instability before performance thresholds are reached. The core bottleneck is therefore not only heating; it is maintaining bounded field geometry and transport suppression long enough to realize net gain and safe conversion.

2. Physical Background

2.1 Aneutronic advantage and direct conversion

Where $D-T$ releases much of its energy in uncharged neutrons (driving materials damage and thermal conversion), $D + He-3$ and $He-3 + He-3$ shift a larger fraction into charged products. Those charged products can be routed through electromagnetic fields for direct conversion. The system-level promise is higher conversion efficiency and reduced activation, conditional on stable operation and safe exhaust handling.

2.2 The instability bottleneck at He-3 regimes

He-3 reactions increase the Coulomb barrier requirements and therefore demand higher temperature and longer confinement. At these regimes, turbulence-driven transport, edge-localized activity, drift waves, and MHD instabilities degrade confinement rapidly. A credible lunar He-3 strategy must therefore treat instability suppression and heat-risk management as first-class objectives, not as side-effects of hardware scaling.

3. CVP Confinement Layer for Lunar He-3 Fusion

3.1 Canonical equation governance (locked)

3.2 Invariant-driven control metrics

CVP operationalizes control through an invariant-driven state vector $I(t)$ constructed from diagnostics and model-derived features. This reduces dependence on brittle single-scalar proxies and improves robustness under drift and configuration changes.

3.3 Curvature Response Operator as actuation design

The Confinement Layer applies engineered actuation families—radial $E \times B$ shear bias waveforms and auxiliary modulation—to damp instability by shaping (i) temporal curvature variation ($\partial K / \partial t$), (ii) spatial curvature gradients (∇K), and (iii) curvature closure/smoothing ($\nabla^2 K$) at the confinement boundary. The near-term claim is conservative: measurable reduction in turbulence power and improved confinement proxies at fixed operating points.

4. Lunar System Architecture: ISRU-to-Reactor-to-Conversion

A lunar He-3 program can be decomposed into three coupled systems: (1) ISRU extraction and refinement of He-3 from regolith, (2) fusion reactor confinement and stability control, and (3) direct energy conversion and charged-particle exhaust management. CVP is primarily a control and governance layer that reduces risk in (2) and improves conversion viability in (3) via constraint enforcement.

4.1 Regolith as a micro-curvature boundary (analogy to confinement)

The lunar regolith is jagged and porous at microscopic scales. In CVP terms, it behaves as a micro-curvature boundary that strongly attenuates electromagnetic propagation. The same language—bounded propagation through structured geometry—can be used to reason about plasma boundary conditioning, where micro-structure and edge gradients drive losses.

5. TEVR Proof Bundles and Acceptance Gates

Because fusion control results are vulnerable to drift, tuning artifacts, and shot-to-shot variation, CVP requires TEVR governance. Any claimed improvement is credible only if it can be replayed from a deterministic proof bundle under declared tolerance ϵ .

5.1 Acceptance gates

6. Roadmap

Phase A: Modify RF magnetized plasma testbed to emulate high-temperature turbulence thresholds relevant to He-3 regimes; demonstrate control authority (P_i down, τ_{up}).

Phase B: Advance to a platform where charged-particle yield proxy (or neutron-visible proxy) can be measured; validate repeatability and constraint compliance.

Phase C: Integrate with partner device control stack as an overlay safety and optimization layer; validate scaling KPIs and multi-day reproducibility.

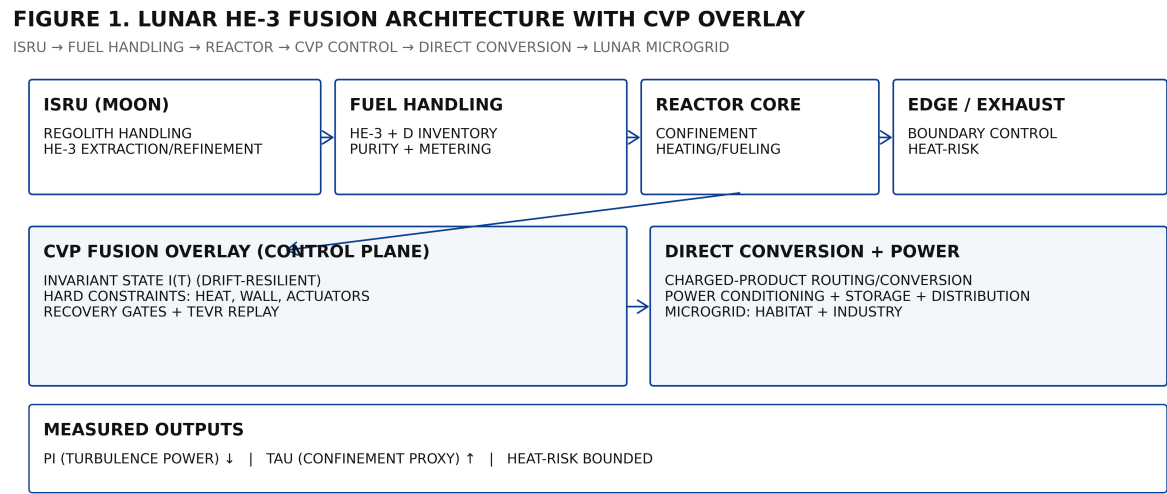
7. Conclusion

Lunar He-3 fusion is compelling largely because it makes direct electromagnetic energy conversion plausible. But the path runs directly through instability suppression at extreme regimes. CVP addresses this by treating confinement as a governed control problem: invariant state, curvature-response actuation

families, strict local recovery, and TEVR proof bundles. The near-term deliverable is not a claim of net gain, but a measured, repeatable reduction in instability signatures and improved confinement proxies that can be staged toward higher-performance platforms.

Figures

Figure 1. Lunar He-3 fusion architecture (ISRU -> reactor -> CVP overlay -> direct conversion -> microgrid).

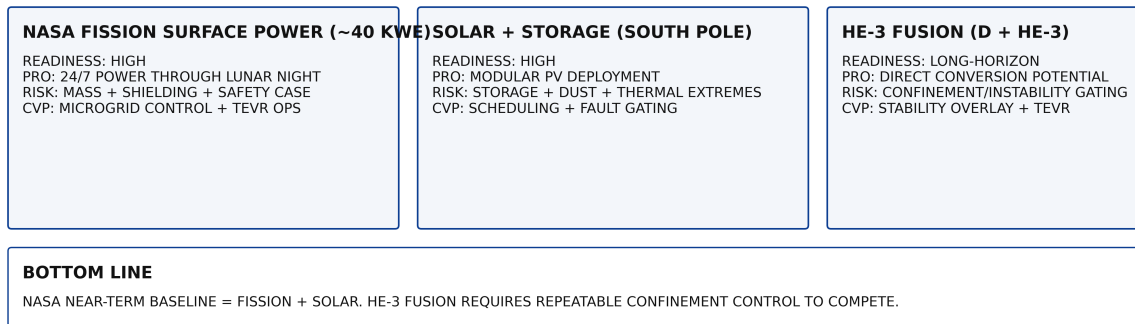


NOTE: CVP OVERLAY IS DEVICE-AGNOSTIC CONTROL + GOVERNANCE (CONSTRAINTS, RECOVERY, REPLAY).

Figure 2. Comparison of lunar power options (NASA fission surface power and solar) vs He-3 fusion and where CVP adds value.

FIGURE 2. LUNAR POWER OPTIONS: NASA BASELINE VS HE-3 FUSION

CVP CONTRIBUTES A SHARED CONTROL/GOVERNANCE LAYER ACROSS OPTIONS



6. Comparison to Current Lunar Power Plans (NASA) and Terrestrial Nuclear Roadmaps

6.1 NASA lunar surface power baseline: fission and solar

NASA's publicly stated lunar power architecture emphasizes fission surface power and solar power as the two primary options for sustained lunar operations, especially at the South Pole where lunar night and thermal extremes challenge photovoltaics. NASA's Fission Surface Power project is targeting a 40-kilowatt-class fission system for lunar use in the early 2030s, and NASA states this effort builds on the earlier Kilopower technology demonstrations.

6.2 Where He-3 fusion differs

Compared to near-term fission surface power, He-3 fusion is a longer-horizon pathway. Its main system-level attraction is the potential for a higher fraction of charged reaction products and direct electromagnetic energy conversion. Its gating risk is confinement: higher operating temperatures and longer confinement times push the system into turbulence and instability regimes. Therefore the primary differentiator for a credible He-3 program is repeatable instability suppression and heat-risk governance, not only ISRU scale.

6.3 Relationship to current terrestrial nuclear discussions (SMRs and microreactors)

Current terrestrial nuclear roadmaps increasingly emphasize small modular reactors (SMRs) and microreactors for scalable, transportable, and faster-deployable generation. These programs reinforce the same credibility requirement: staged demonstrations, hard safety envelopes, and evidence packages that regulators and partners can trust. CVP's TEVR discipline aligns with this movement by making control improvements auditable and replayable, rather than anecdotal.

References (selected, public)

NASA Fission Surface Power:
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NASA Kilopower overview:
<https://www.nasa.gov/directorates/stmd/tech-demo-missions-program/kilopower-hmqzw/>

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Integrated Lunar Power Strategy Considerations (PDF):
https://www.lpi.usra.edu/lunar/strategies/resources/M2M_ACR2025_LunarPowerStrategy.pdf

DOE Advanced SMRs: <https://www.energy.gov/ne/advanced-small-modular-reactors-smrs>

DOE Microreactor Program: <https://gain.inl.gov/doe-microreactor-program/>

Figures (Updated)

Figure 1 (Updated). Lunar He-3 fusion architecture (ISRU -> reactor -> CVP overlay -> direct conversion -> microgrid).

Figure 2 (Updated). Comparison of lunar power options (NASA fission surface power and solar) vs He-3 fusion and where CVP adds value.

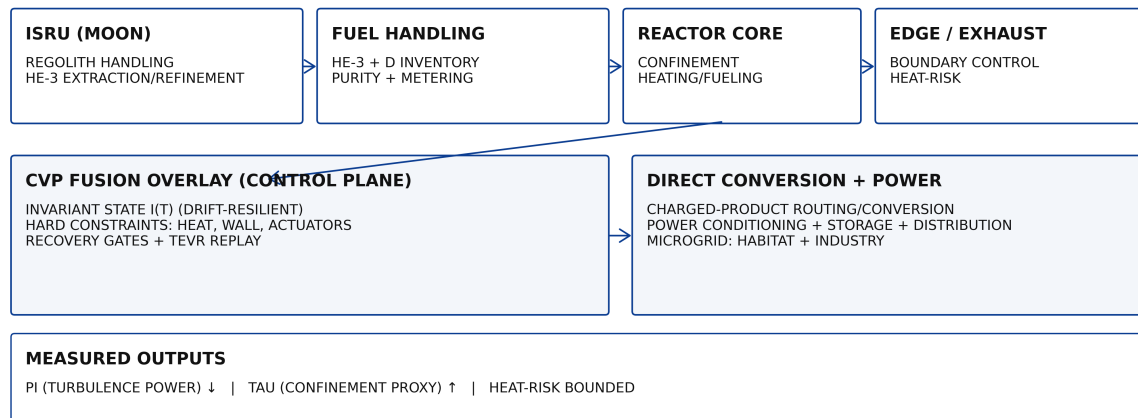
Figure 3 (New). CVP Overlay applied to NASA's lunar baseline power plan (FSP + microgrid operations).

NASA-Style Figures

Figure 1. Lunar He-3 fusion architecture with CVP overlay (NASA-style).

FIGURE 1. LUNAR HE-3 FUSION ARCHITECTURE WITH CVP OVERLAY

ISRU → FUEL HANDLING → REACTOR → CVP CONTROL → DIRECT CONVERSION → LUNAR MICROGRID



NOTE: CVP OVERLAY IS DEVICE-AGNOSTIC CONTROL + GOVERNANCE (CONSTRAINTS, RECOVERY, REPLAY).

Figure 2. Lunar power options comparison (NASA baseline vs He-3 fusion; NASA-style).

FIGURE 2. LUNAR POWER OPTIONS: NASA BASELINE VS HE-3 FUSION

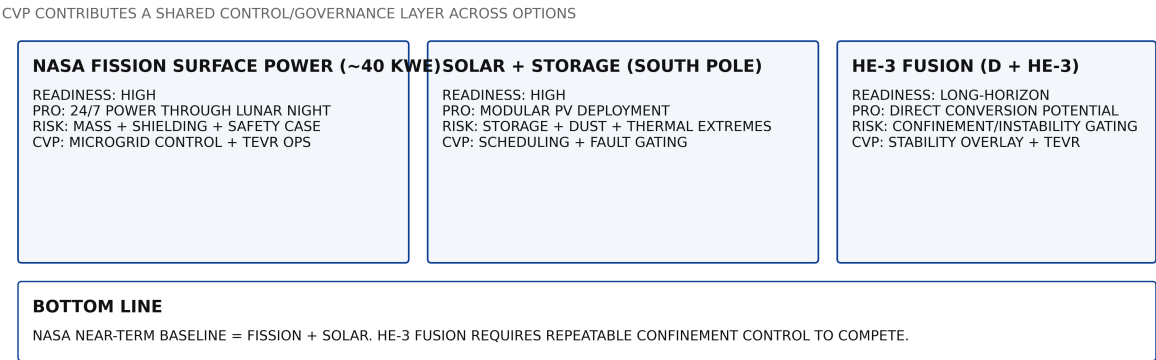
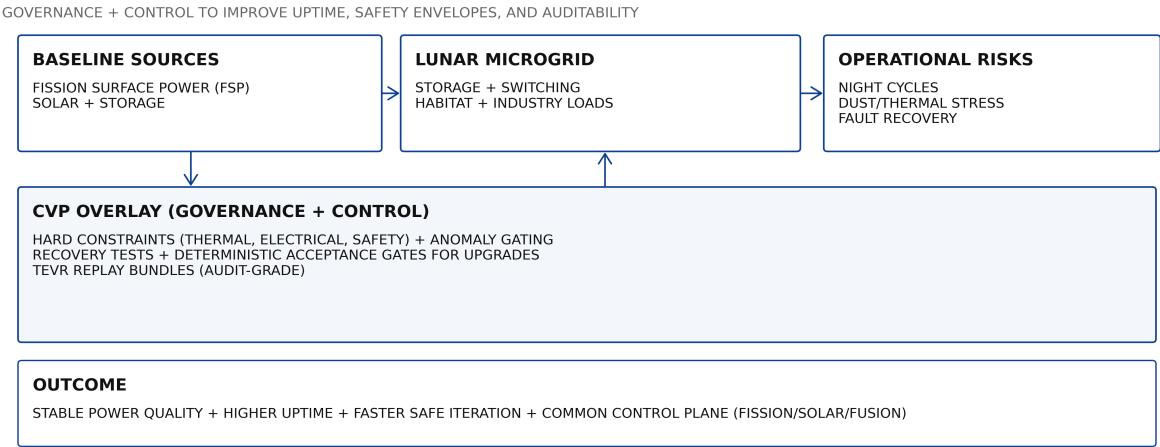


Figure 3. CVP overlay applied to NASA baseline lunar power operations (NASA-style).

FIGURE 3. CVP OVERLAY ON NASA BASELINE LUNAR POWER OPERATIONS

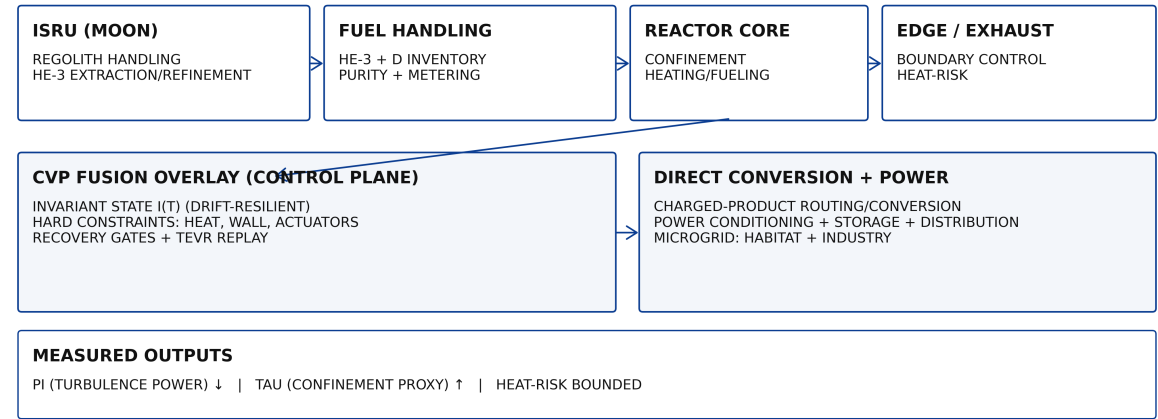


NASA-STYLE ARCHETYPE FIGURES (V2)

Figure 1. Lunar He-3 fusion architecture with CVP overlay (NASA archetype).

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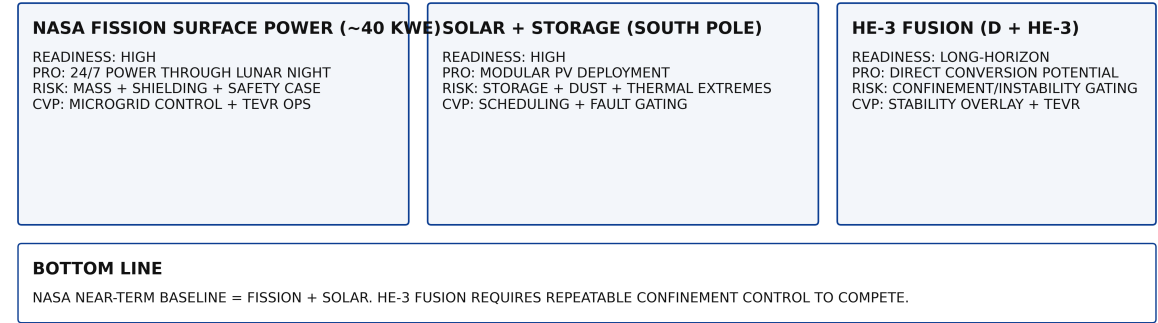


Figure 3. CVP overlay on NASA baseline lunar power operations (NASA archetype).

FIGURE 3. CVP OVERLAY ON NASA BASELINE LUNAR POWER OPERATIONS

GOVERNANCE + CONTROL TO IMPROVE UPTIME, SAFETY ENVELOPES, AND AUDITABILITY

